

RAFFLES INSTITUTION
2018 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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BIOLOGY

Paper 4 Practical

9744/04

24th August 2018

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen on both sides of the paper.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **questions 1 and 2** in the **spaces** provided on the **Question Paper**.

Answer **question 3** in the **foolscap paper** provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **11** printed pages.



Raffles Institution
Internal Examination

Answer all the questions.

- 1 Glucose in solution reduces pink, acidified potassium manganate (VII) solution, **P**, to a colourless or slightly yellow solution at end-point.

The rate of the colour change depends on the concentration of the glucose solution. The greater the concentration of glucose solution, the faster the end-point is reached.

You are required to investigate the rate of colour change in five glucose solutions of different concentrations, **W1** to **W5**, by finding the time taken to reach the end-point.

You are provided with the following:

30cm³ of glucose solutions, **W1** to **W5**,

60cm³ of 1 mol dm⁻³ sulfuric acid, labelled **S**,

30cm³ of 0.04% potassium manganate (VII) solution, labelled **P**.

Sulfuric acid, S, and potassium manganate (VII) solution, P, are harmful. You are advised to wear the eye protection and gloves you have been given and wash off any spillages.

The 10cm³ syringe will need to be re-used with different solutions. Before re-using, wash the syringe thoroughly using distilled water provided, labelled **washing water**. Flush distilled water through the syringe and discard the waste into the container labelled **waste**.

Proceed as follows.

Stage 1

Label five boiling tubes, **W1**, **W2**, **W3**, **W4** and **W5**.

Stage 2

Put 10cm³ of each glucose solution, **W1**, **W2**, **W3**, **W4** and **W5**, into the correspondingly labelled boiling tube.

Put 5cm³ of solution **S** into each boiling tube.

Insert the rubber bung into boiling tube **W1** and, with your finger holding the rubber bung in place, gently mix the solution in the boiling tube. Do not turn the boiling tube upside-down. Repeat for boiling tubes **W2**, **W3**, **W4** and **W5**.

Stage 3

Start the stopwatch. The stopwatch should run continuously throughout the whole investigation. Do not stop the stopwatch until all start and end-point times for all glucose solutions have been recorded in stages **4**, **5**, **6**, and **7**.

Stage 4

Put 2cm³ of solution **P** in the boiling tube labelled **W1**. Immediately insert the rubber bung, mix well and record the start time in Table 1.1 on page 3. *Record your readings in the following format, mm:ss, e.g. 2 minutes 15 seconds is recorded as 02:15.*

Stage 5

Repeat stage 4 with boiling tube **W2**, **W3**, **W4** and **W5**. Record the start times in Table 1.1.

Stage 6

- (a) Observe the boiling tubes **W1**, **W2**, **W3**, **W4** and **W5** and record, in Table 1.1, the time when each end-point is reached, using appropriate format. These are the raw data for Trial 1.

If the time taken to reach the end-point for any boiling tube is longer than 300 seconds, record 'no change'. Complete the table to show the time taken, in seconds, to reach each end-point.

Table 1.1

	Boiling tube				
	W1	W2	W3	W4	W5
Start time					
End time					
Time taken to reach end-point /s					

Stage 7

- (b) In order to increase confidence in the data, repeat stages 1 to 6.

Record your results in Table 1.2, using appropriate format. These are the raw data for Trial 2. Complete the table to show the time taken, in seconds, to reach each end-point.

Table 1.2

	Boiling tube				
	W1	W2	W3	W4	W5
Start time					
End time					
Time taken to reach end-point /s					

[4]

- (c) Prepare and complete a table in the space below to show the mean time taken to reach the end-point for each glucose solution.

[4]

- (d) A sixth boiling tube should be set-up as a control. Describe how a control is set-up and explain the reason for this.

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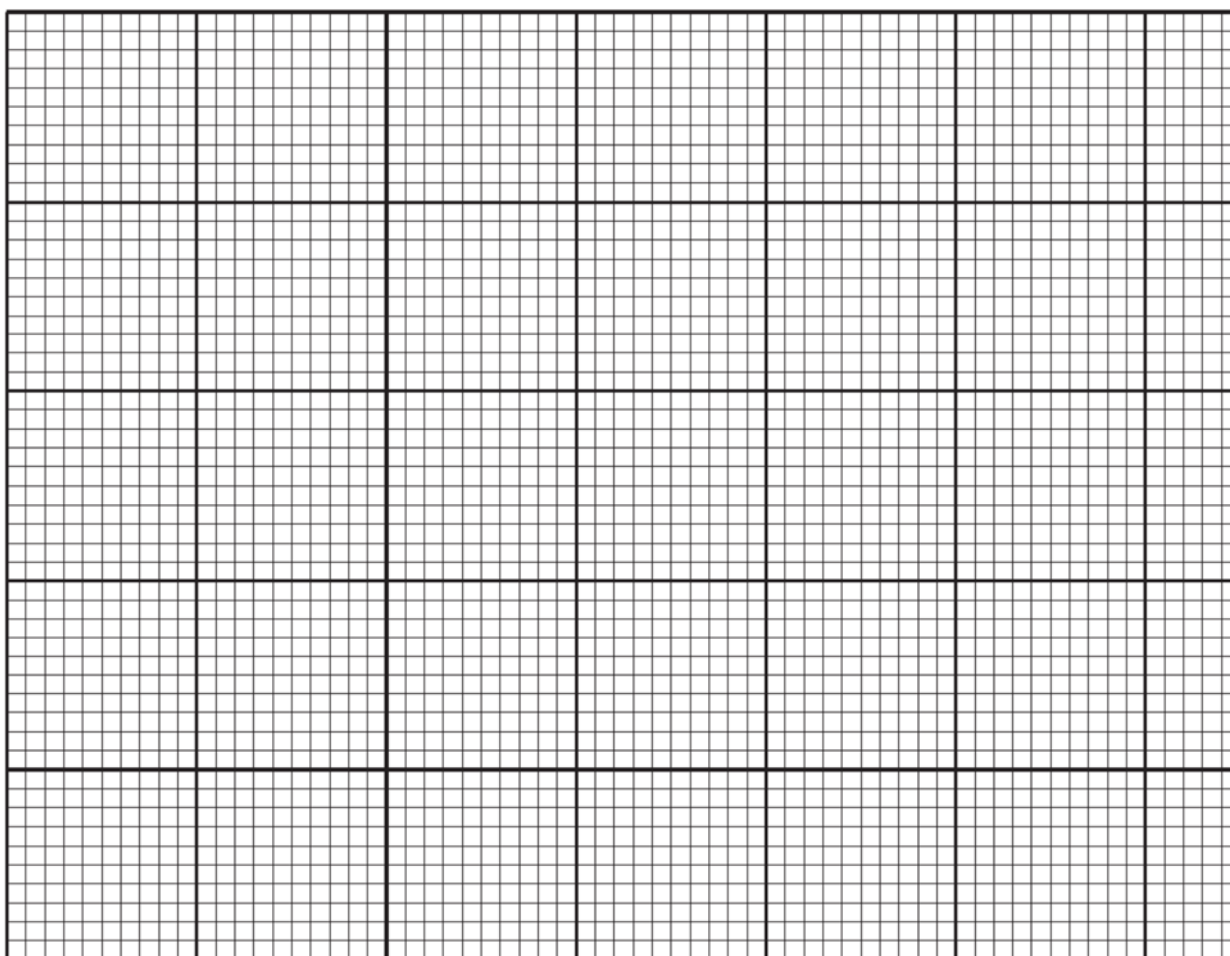
.....

.....[2]

- (e) A student proceeded to repeat the experiment with known glucose concentrations. He recorded his results as shown below.

Glucose concentration / %	Time taken to reach end-point / s
5	115
10	88
15	62
20	36
25	10

Present the above data in a graphical format.



[5]

- (f) (i) Using the graph, estimate the concentration of solution **W1**.

Concentration of solution **W1** :[1]

- (ii) Without extrapolating the graph, estimate the concentration of solution **W5** using the formula $y = mx + c$.

Show your working clearly, including how the values of m and c are calculated.

Concentration of solution **W5** :[3]

[Total: 19]

- 2 You are provided with a prepared slide showing a section of a root tip with cells at various stages of mitosis.
- (a) Examine the slide carefully. Identify **2** cells at different stages of mitosis and make a large labelled drawing of **each** cell. Indicate the stage of mitosis for each of the cells drawn.

[6]

A student viewed the slide under a microscope with a high-powered objective lens and measured the length of a chromosome using the eyepiece graticule.

Fig. 2.1 shows the view that the student saw when looking down the microscope at the eyepiece graticule and the chromosome.

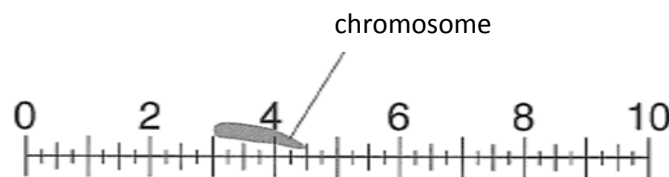


Fig. 2.1

Using a stage micrometer with 0.01mm divisions, the students found that the entire length of the eyepiece graticule on Fig. 2.1 matched 2 divisions on the stage micrometer.

- (b) Calculate the actual length of the chromosome shown in Fig. 2.1 in μm .

..... μm [3]

- (c) Using the slide provided, select a region with 100 actively dividing cells. As a guide, the region should contain cells in at least 2 different stages of the mitotic cell cycle. Count the number of cells in each stage of the mitotic cell cycle. Repeat the count for another chosen field of view.
- (i) Prepare and complete a table in the space below to show the percentage of cells in each stage of the mitotic cell cycle.

[5]

- (b) The percentage of cells observed in each stage can be used to infer the duration of the stage.

- (i) State the longest stage of the cell cycle and explain how you arrived at your conclusion.

.....

.....

.....

..... [2]

- (ii) A student commented that an additional piece of information is required to estimate the duration of each stage. State what is this piece of information.

.....

..... [1]

- (iii) Another student commented that this method of estimating the duration of each stage has a huge margin of error.

Other than the absence of 3 replicates, suggest why it is so and describe how you can improve on the accuracy of the results obtained.

.....

 [2]

- (c) In another experiment, the effect of temperature on cell division in root tips was investigated.

Samples of root were grown at **T1** (20°C) and **T2** (40°C). 200 cells were randomly selected and the number of dividing cells was recorded in Table 2.1.

The number of dividing cells in a random selection of 200	
T1 (20°C)	T2 (40°C)
22	69
46	48
35	58
33	53
20	67
28	42
37	72
31	57
39	68
23	49

Table 2.1

A *t*-test was used to find out about the nature of the difference in the number of dividing cells at **T1** and **T2**. The calculated *t*-value is 6.423.

$$v = n_1 + n_2 - 2$$

Using the formula and the probability table provided, state the conclusion can be drawn from this *t*-test.

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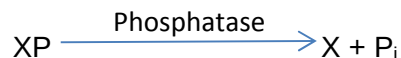
 [3]

Degrees of freedom	Significance level					
	20% (0.20)	10% (0.10)	5% (0.05)	2% (0.02)	1% (0.01)	0.1% (0.001)
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.598
3	1.638	2.353	3.182	4.541	5.841	12.941
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.859
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.405
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.767
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.043	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.158	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

[Total: 22]

- 3 Phosphatases are enzymes that catalyse the removal of phosphate groups from certain types of organic phosphate.

The principle of the reaction involving phenolphthalein diphosphate (**XP**) catalysed by phosphatase is as follows.



When colourless phenolphthalein diphosphate (XP) is catalysed by phosphatase enzyme, phenolphthalein (X) is released. Phenolphthalein then reacts with sodium carbonate giving a pink colouration. The colour intensity can be measured using a spectrophotometer and readings are obtained in absorbance value/A.

It is known that **Q**, an organic compound, is an inhibitor of phosphatase. Using the list of apparatus below, design an experiment to determine if **Q** is a competitive or a non-competitive inhibitor.

Reagents and apparatus

- 10% Solution of **XP**
- 5% Phosphatase solution
- 5% Inhibitor solution **Q**
- 5% sodium carbonate solution
- pH buffer (pH 7)
- Distilled water
- 10 cm³, 5 cm³ and 1 cm³ syringes
- Glass rods
- Test tubes and rack
- Cuvettes
- Spectrophotometer
- Normal laboratory glassware, e.g. beakers, measuring cylinders, graduated pipettes, glass rods, etc.

Your plan should :

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- be illustrated by relevant diagrams, if necessary, to show for example the arrange of the apparatus used.
- identify the appropriate variable.
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible.
- include layout of results tables and graphs with clear headings and labels.
- use the correct technical and scientific terms.
- include reference to safety measures to minimise any risk associated with the proposed experiment.

[Total: 14]

-END-

Aim : To determine if Q is a competitive or a non-competitive inhibitor of phosphatase activity.

Theory

- The substrate, Phenolphthalein diphosphate (XP) complementary in shape and charge to the active site of phosphatase and the phosphatase catalyses the transfer of inorganic phosphate group from XP producing phenolphthalein.
- Phenolphthalein then reacts with sodium carbonate solution to form a pink compound. The intensity of the pink compound can be measured using a spectrophotometer(dependent variable).
- The effects of inhibitor Q on phosphatase activity can be investigated over a range of substrate concentration.
- If Q is a competitive inhibitor that binds to the active site and competes with the substrate for the active site
- The V_{max} should remain the same. The effects of inhibitors will be overcome at higher concentration of substrate.
- If Q is a non-competitive inhibitor that binds to a site other than active site, causing a change in the active site.
- The V_{max} will be lowered. The effects of inhibitors is not overcome at higher concentration of substrate.

[Theory T1, T2, T3, max 3 marks]

Variables

	Description	How it is controlled
Independent Variable	<u>Presence of inhibitor Q</u>	• Conduct experiment, <u>with and without inhibitor Q.</u> Accept • Use various concentration of <u>inhibitor Q of 1,2,3,4 and 5% (5 regular concentration of inhibitor Q).</u>
Dependent Variable	<u>Rate of phosphatase activity</u>	• obtained by measuring <u>intensity of pink colouration using a spectrophotometer</u>

[Independent Variable and how it is controlled - IDV]

[Dependent variable and how it is measured – DV]

Procedure

Pilot test

Conduct a pilot experiment to determine suitability of apparatus, range of independent variables (e.g. range of concentration of inhibitor Q), duration of experiment & amount of materials used.

Numbered stepsPreparation of inhibitor solution

- Using 10cm³ syringe, prepare 5 different concentrations of XP solution by following the dilution table below: **(independent variable)**

Concentration of XP solution /%	Volume of 10% XP solution used /cm ³	Volume of distilled water used /cm ³
2	2.0	8.0
4	4.0	6.0
6	6.0	4.0
8	8.0	2.0
10	10.0	0.0

[Dilution table, DT]Preparation of enzyme reaction

- Set up a water bath of 37°C, using hot and cold water, and use a thermometer to monitor temperature. (constant variable, temperature)
- Place inhibitor Q solution, XP solution and phosphatase solution separately into the water bath and allow 1 min of equilibration time, so that solutions can attain temperature of water bath. (Equilibration time, ET)
- Using a syringe add 1 cm³ of 2%, 4%, 6%, 8% and 10% XP solution to labelled test tubes. (range of substrate concentration)
- Add 1 cm³ of 5% inhibitor Q solution to each test tube (independent variable, inhibitor Q)
- Add 1 cm³ of 5% Phosphatase solution to each test tube. Mix well and incubate the mixture in the water bath. (constant variable, concentration of enzyme) (constant variable, volume of enzyme)
- After 5 minutes, (constant variable, reaction time) transfer 1cm³ of reaction mixture (constant variable, volume of sample used to measure A) into a cuvette and measure the absorbance value/colour intensity of the pink solution using a spectrophotometer. (dependent variable, Absorbance of solution)
- Repeat Step 3-8 using distilled water instead of inhibitor solution.
- Record the colour intensity in absorbance value (A) for each concentration of inhibitor Q solution in a table format.
- Plot a graph of colour intensity in absorbance value/A against concentration of inhibitor solution.

-[Controlled variable and how it is controlled CV1, CV2, CV3, max 3 marks]**[Equilibration time, ET]**Control

- Keep all variables constant, set up control experiment using distilled water instead of solution of inhibitor Q.
- This is to obtain baseline results, in the absence of inhibitors.

[Controls and purpose, C]

Repeats & replicates

- Conduct 3 replicates for each condition to check that no anomalies are present
- Repeat entire experiment twice to check for reproducibility.

[Repeat and replicates, RR]

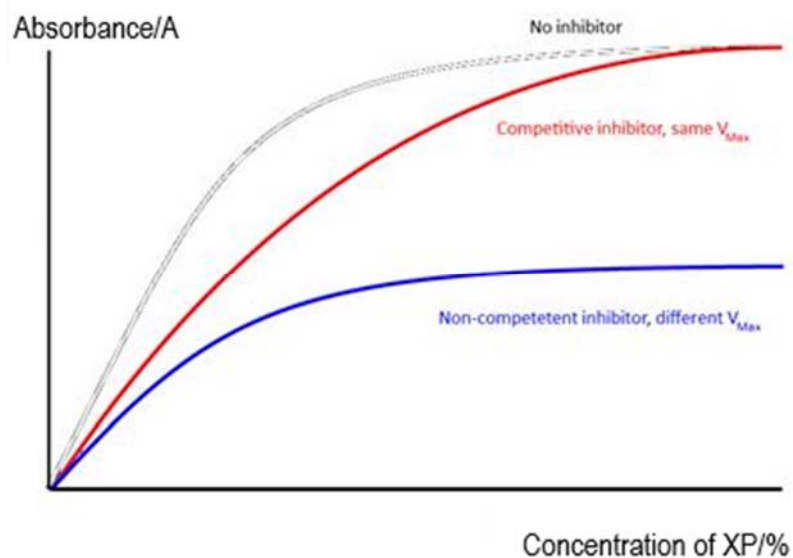
Data Recording and Processing

Record and process data as follows: (table and graph should have same heading / unit)

Concentration of XP /%	Absorbance/A							
	With inhibitor Q				Without inhibitor			
	Trial 1	Trial 2	Trial 3	Average	Trial 1	Trial 2	Trial 3	Average
2								
4								
6								
8								
10								

[Data table : DT]

Graph showing effect of inhibitor Q on the rate of phosphatase activity at various concentration of substrate



[Expected graphs :G]

[Explanation of how to interpret graph to draw conclusion : I]

Risks & precautions

1. Wear gloves and goggles when handling XP solution, inhibitor Q solution and phosphatase solutions as they may be toxic.

2. Do not touch electrical switch of the spectrophotometer with wet hands to prevent electrocution.
3. Wear oven mittens / insulating gloves when handling hot water to avoid scalding.

[Description of Risk and Precautions]